

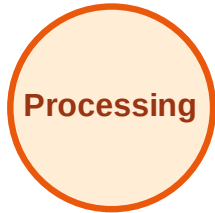
BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Complete



Processing



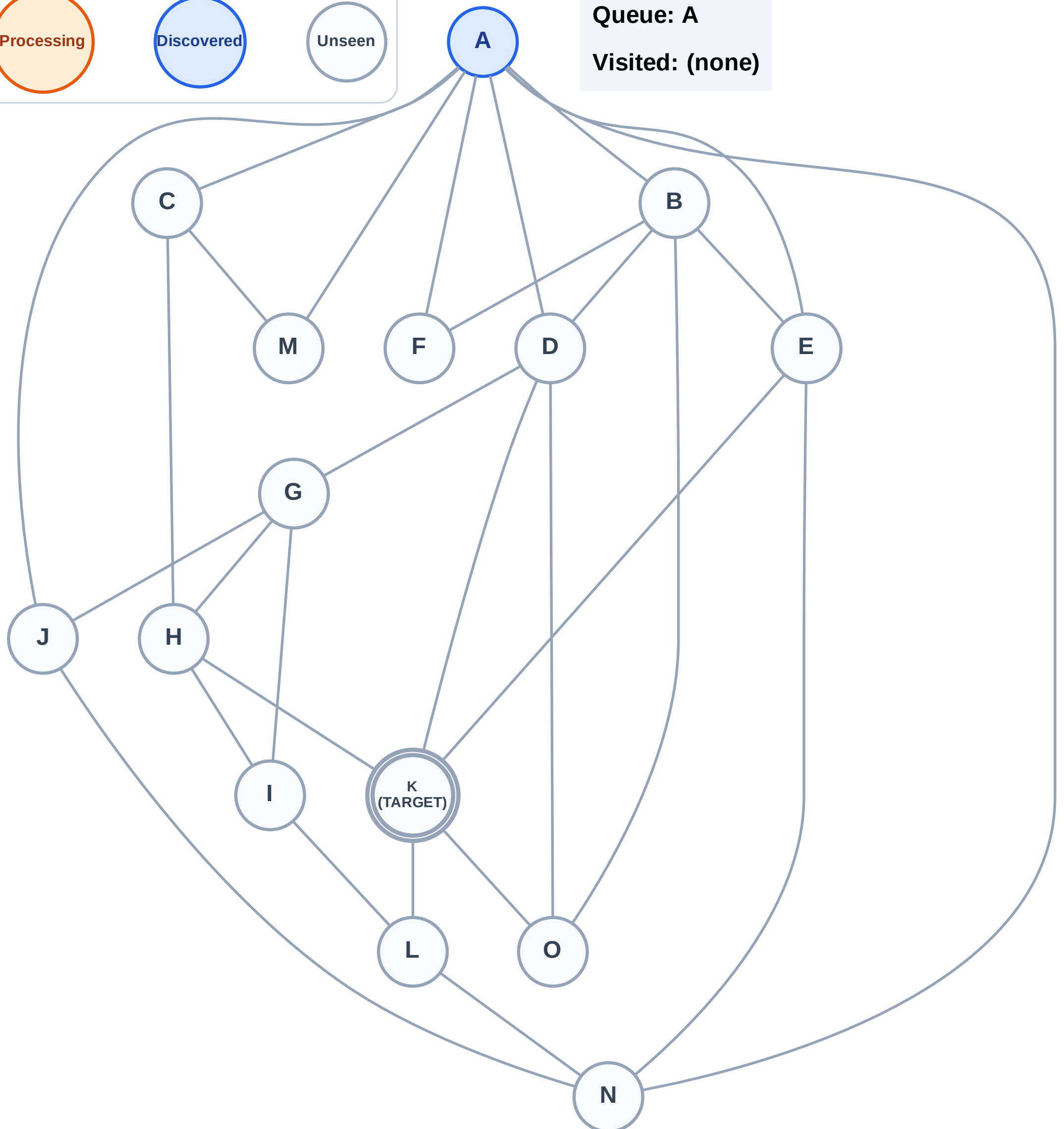
Discovered



Unseen

Queue: A

Visited: (none)



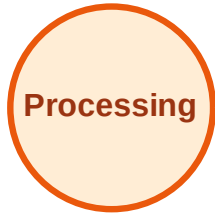
BFS Traversal

Step 2: Process node A

Legend



Complete



Processing



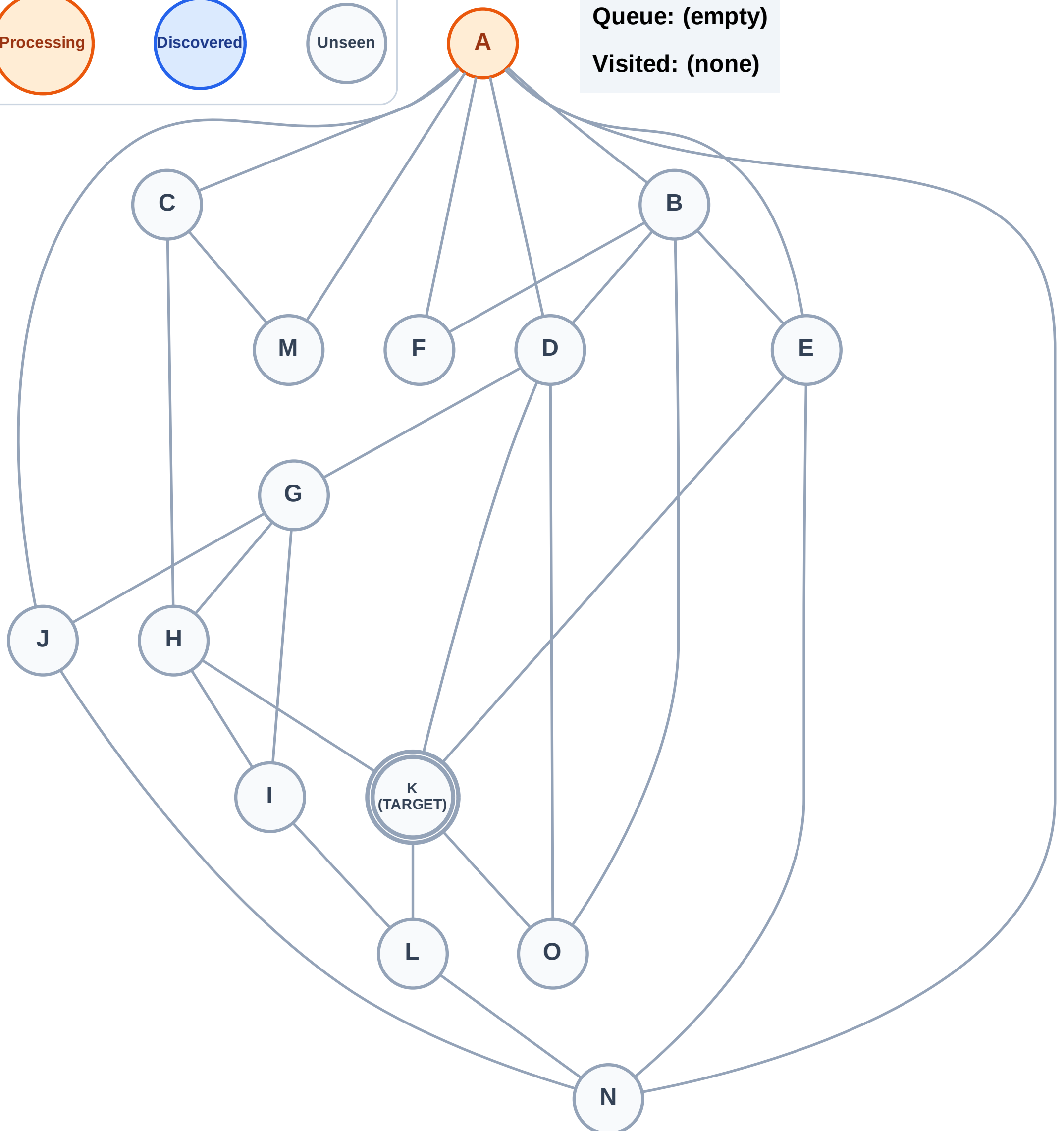
Discovered



Unseen

Queue: (empty)

Visited: (none)



BFS Traversal

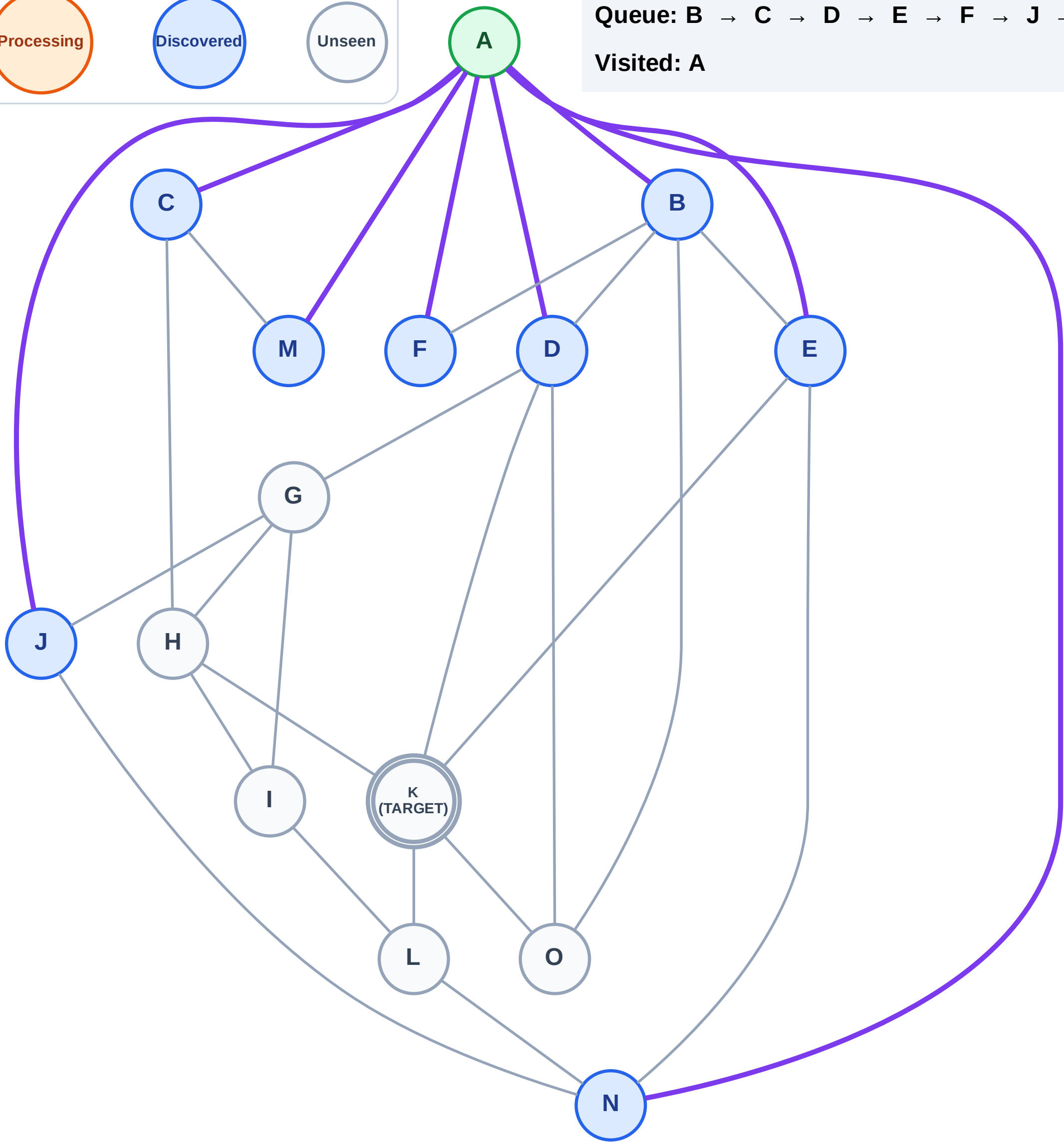
Step 3: Discover B, C, D, E, F, J, M, N from A

Legend



Queue: B → C → D → E → F → J → M → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

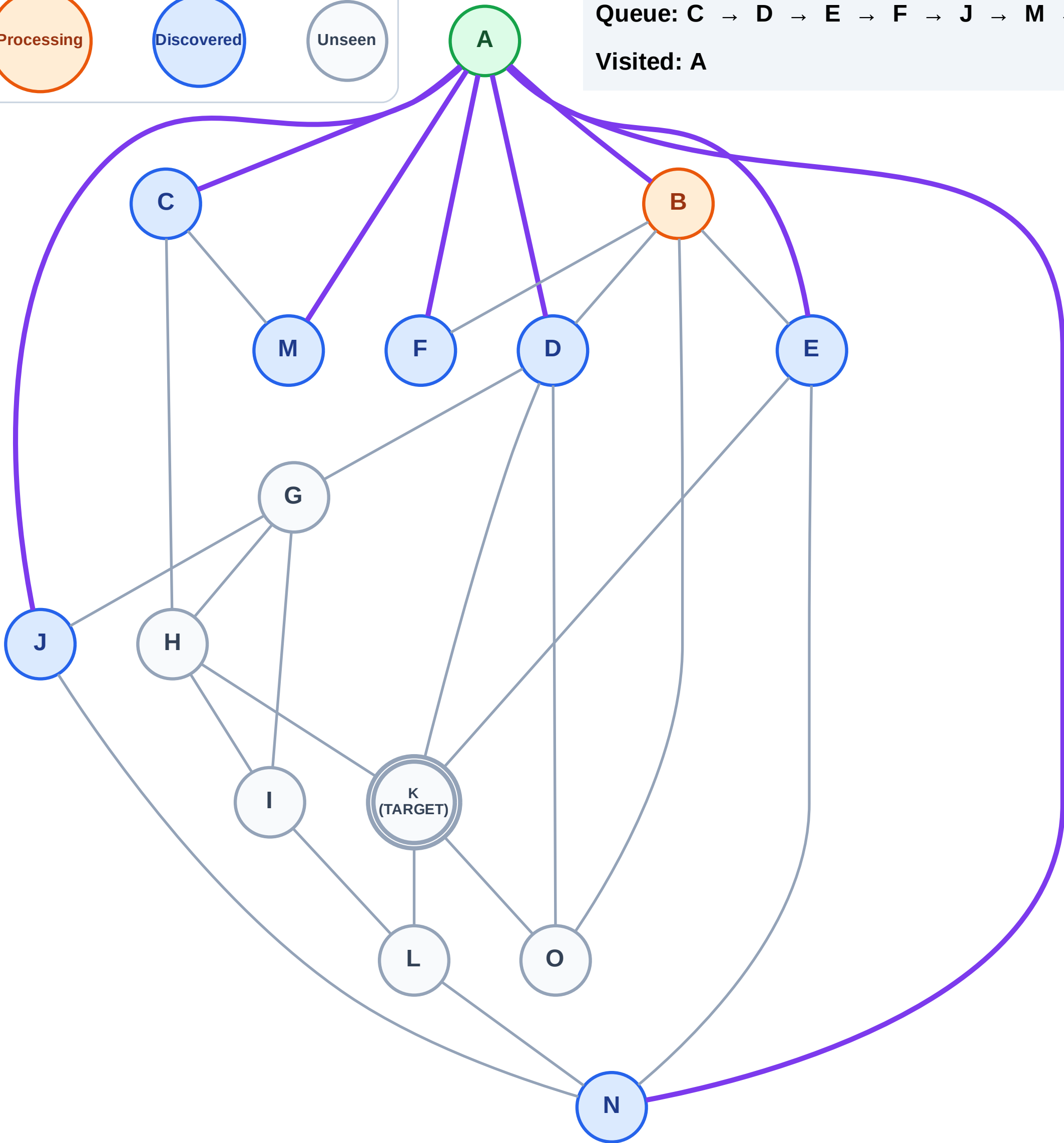
Processing

Discovered

Unseen

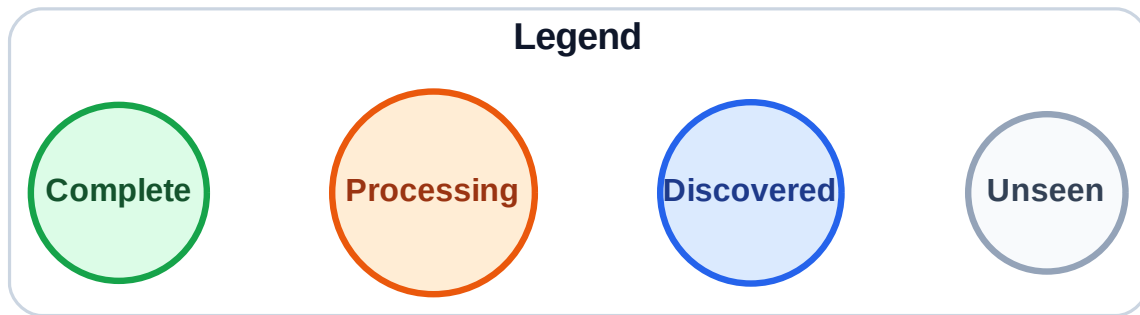
Queue: C → D → E → F → J → M → N

Visited: A



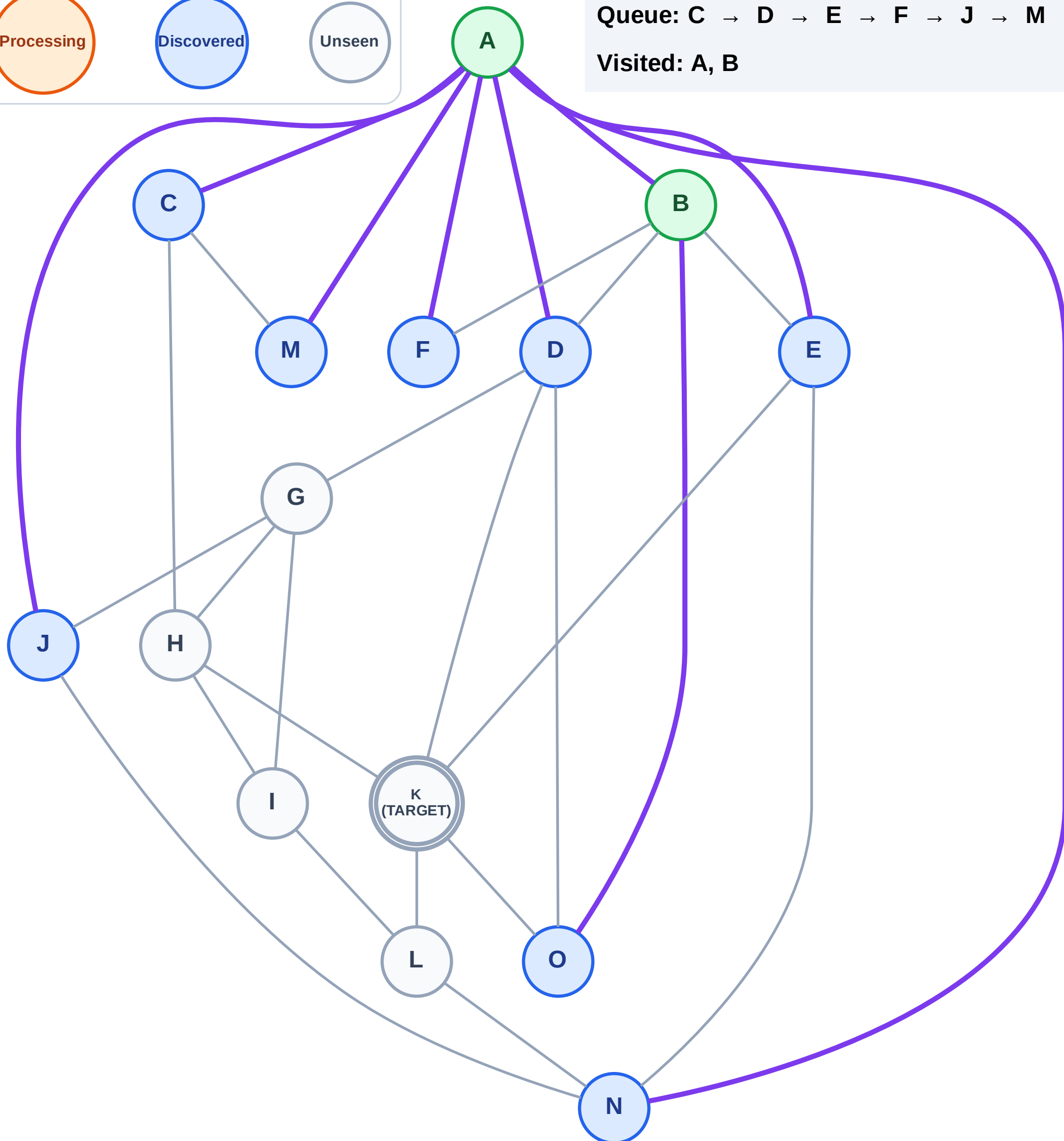
BFS Traversal

Step 5: Discover O from B



Queue: C → D → E → F → J → M → N → O

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

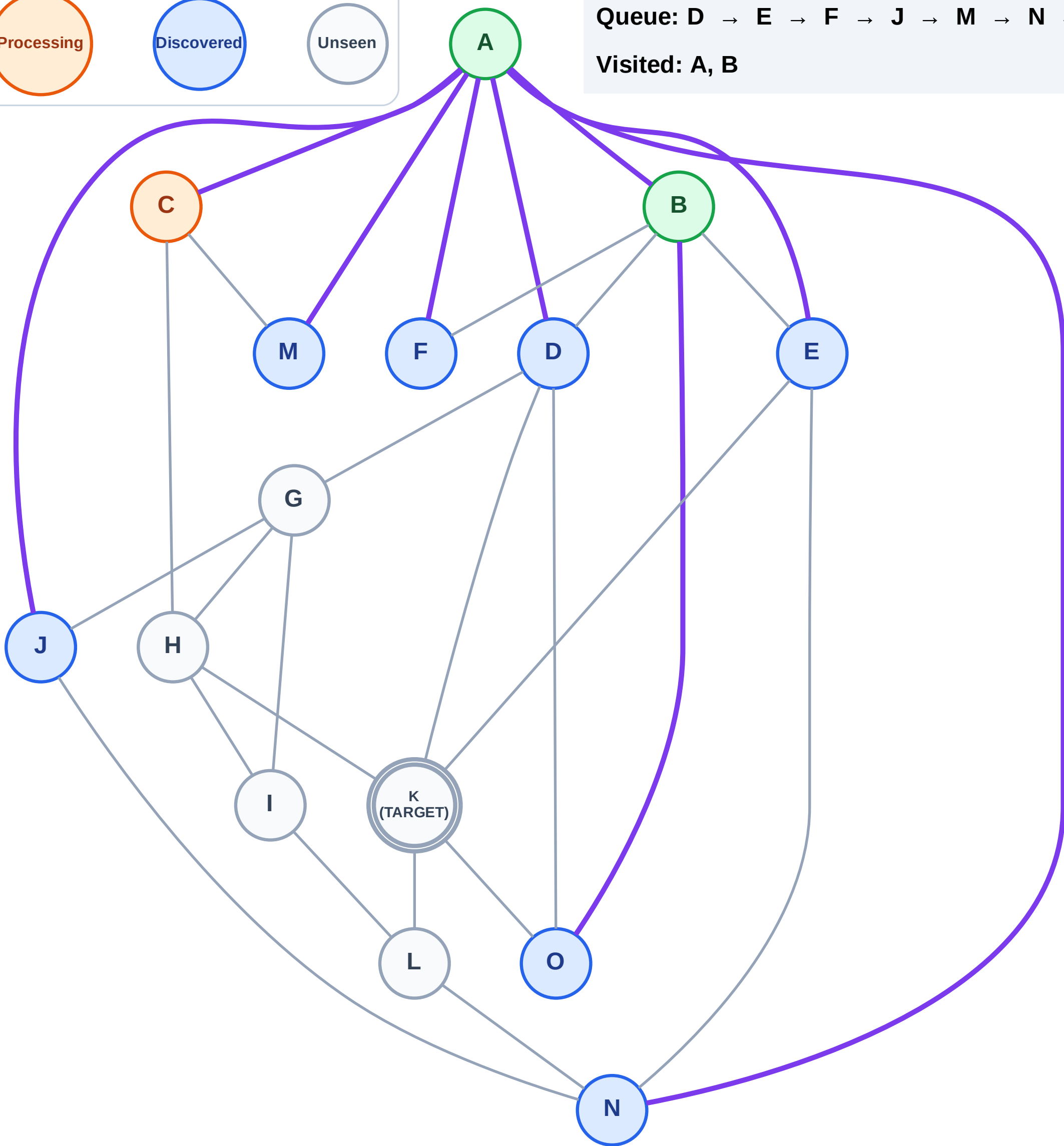
Processing

Discovered

Unseen

Queue: D → E → F → J → M → N → O

Visited: A, B



BFS Traversal

Step 7: Discover H from C

Legend

Complete

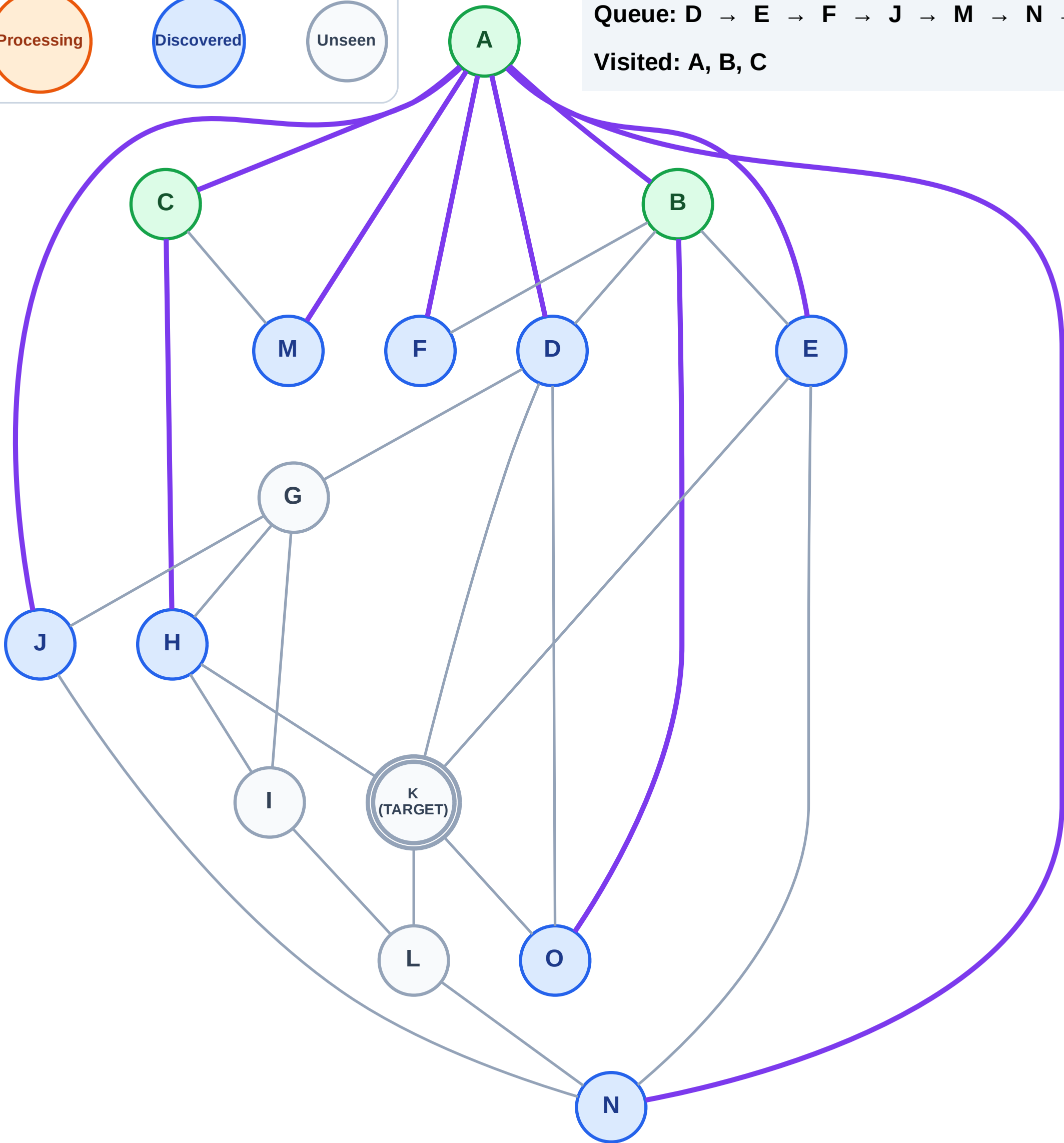
Processing

Discovered

Unseen

Queue: D → E → F → J → M → N → O → H

Visited: A, B, C



BFS Traversal

Step 8: Process node D

Legend

Complete

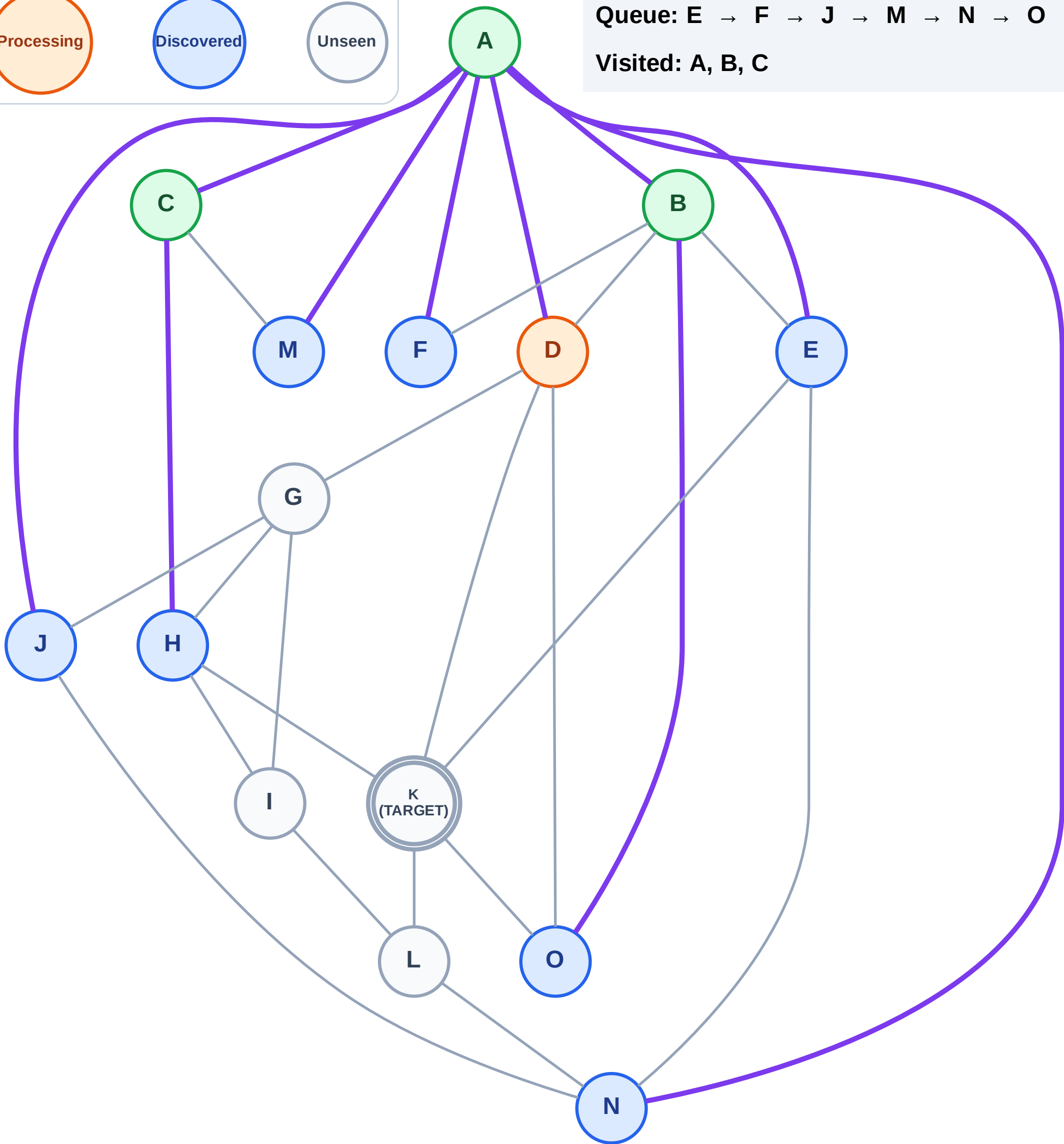
Processing

Discovered

Unseen

Queue: E → F → J → M → N → O → H

Visited: A, B, C



BFS Traversal

Step 9: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

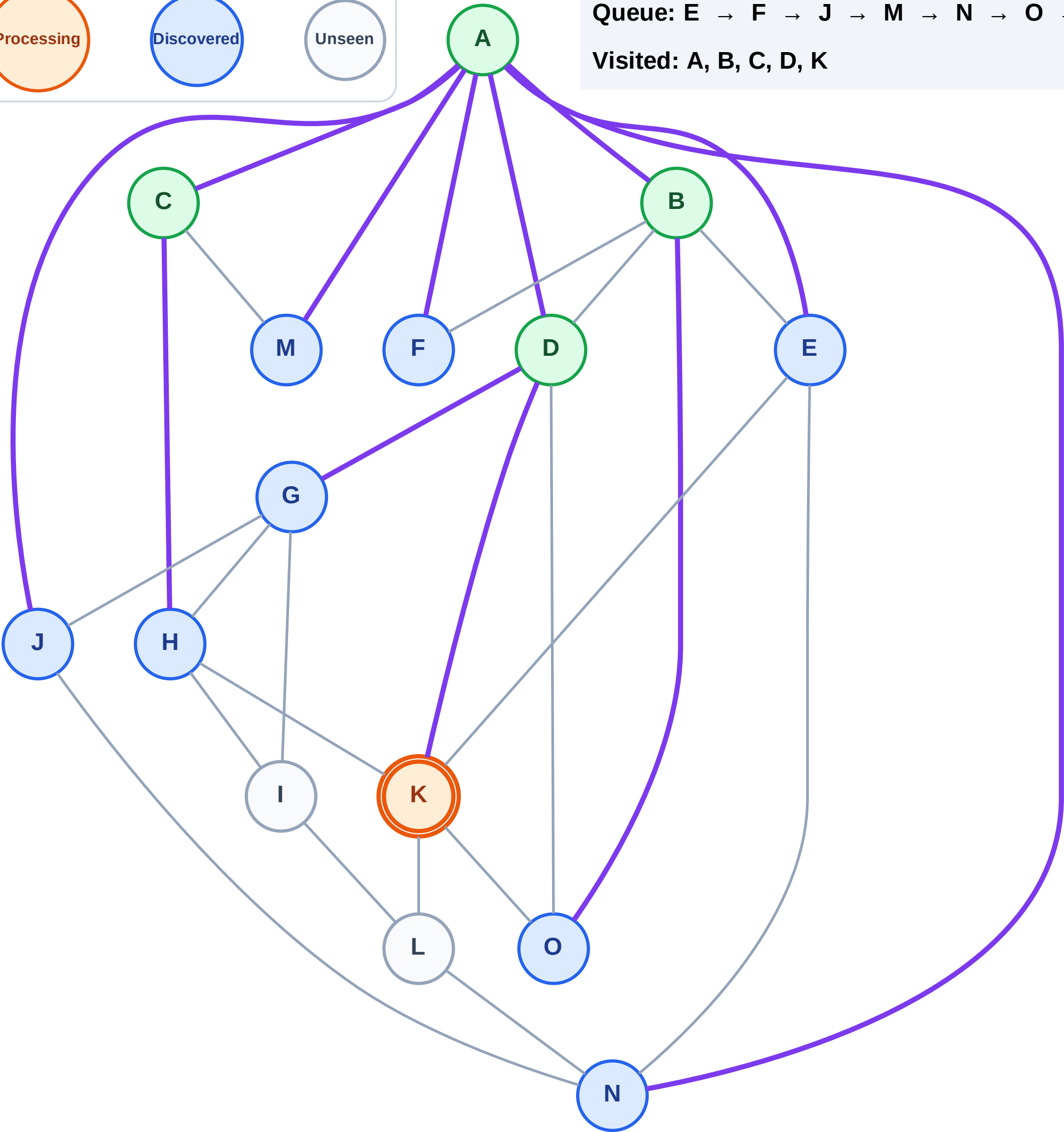
Processing

Discovered

Unseen

Queue: E → F → J → M → N → O → H → G

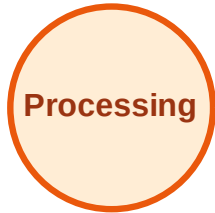
Visited: A, B, C, D, K



DFS Traversal

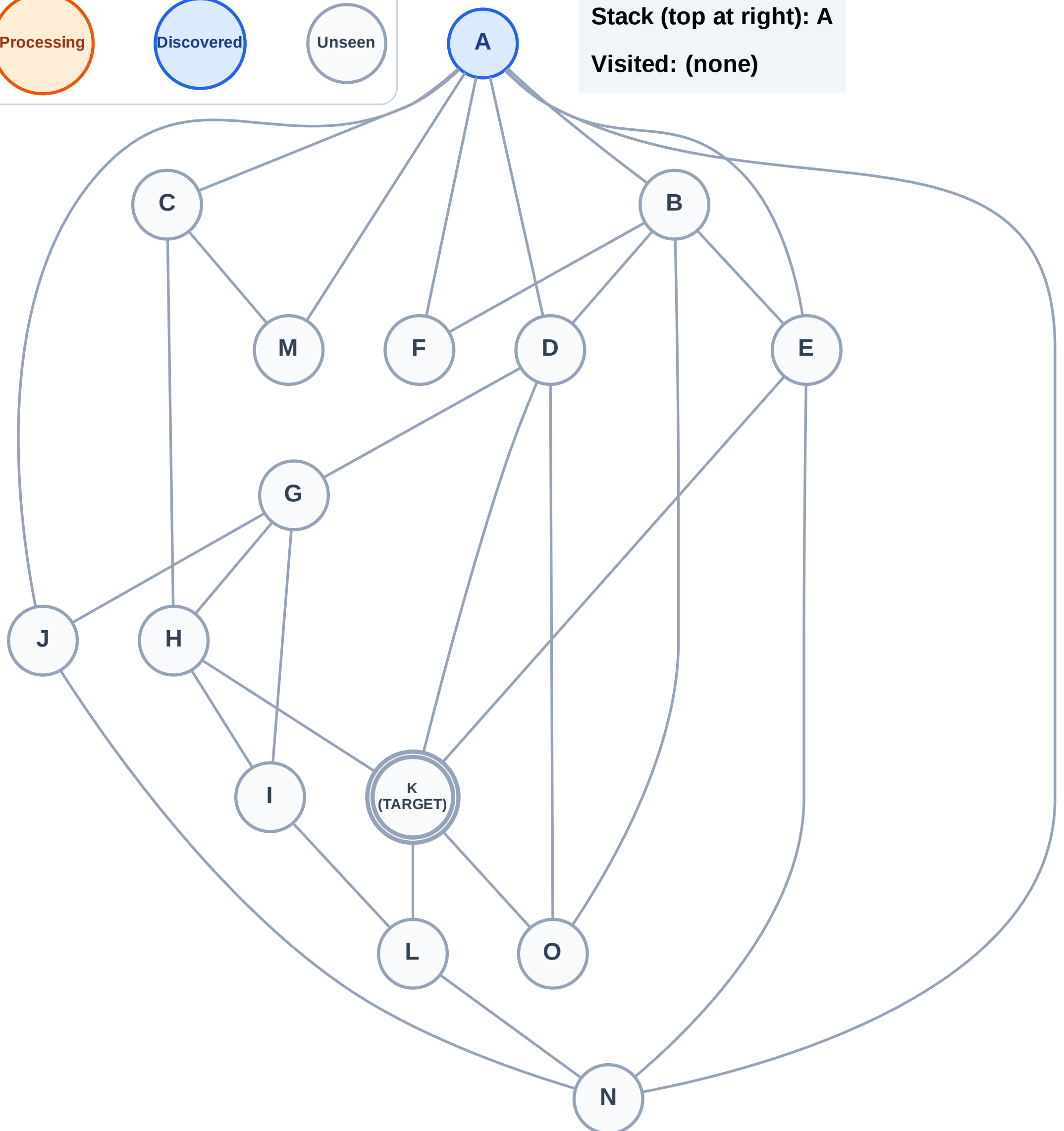
Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend



Stack (top at right): A

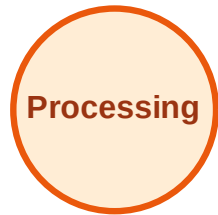
Visited: (none)



DFS Traversal

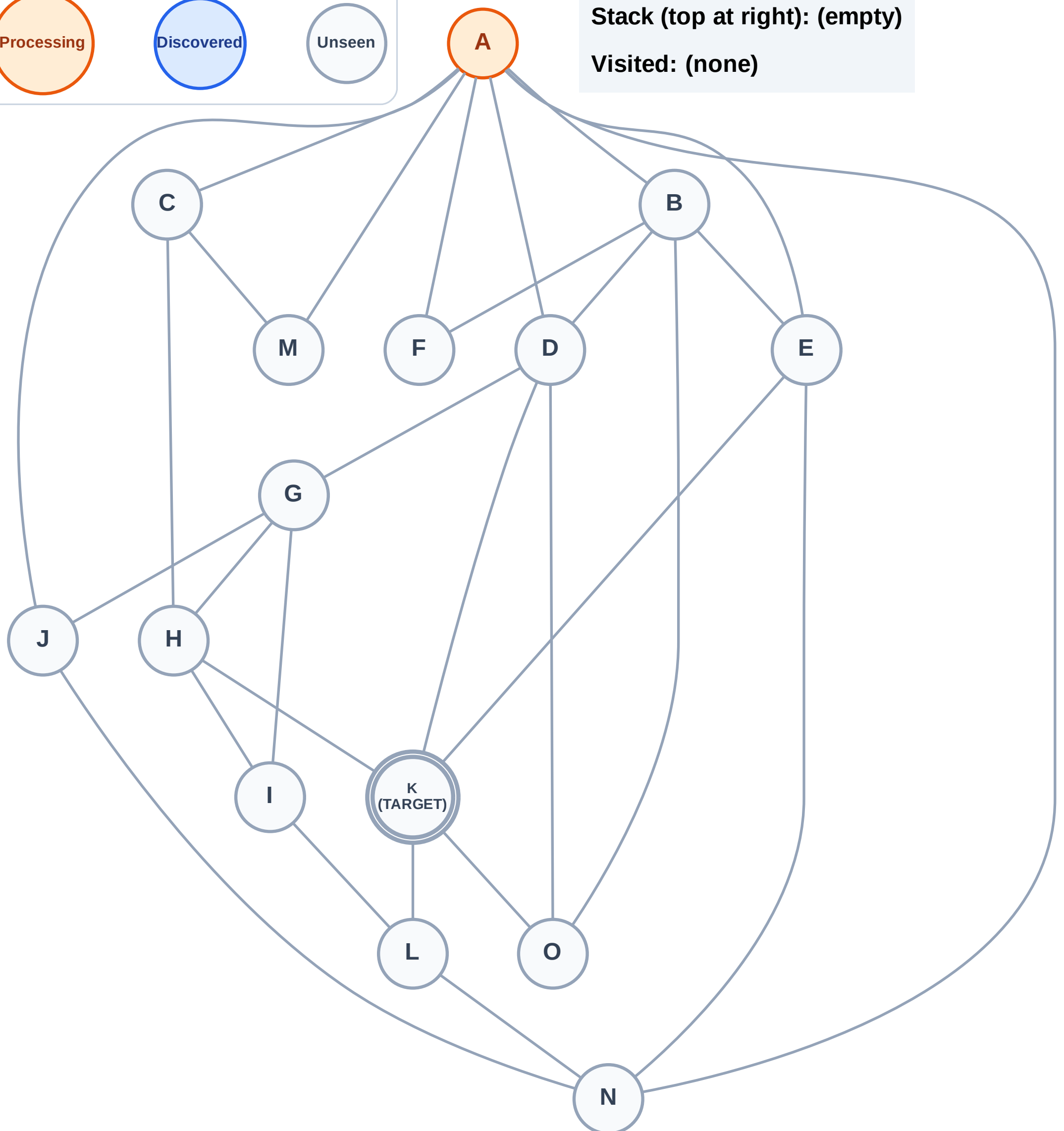
Step 2: Process node A

Legend



Stack (top at right): (empty)

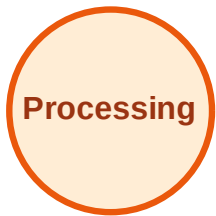
Visited: (none)



DFS Traversal

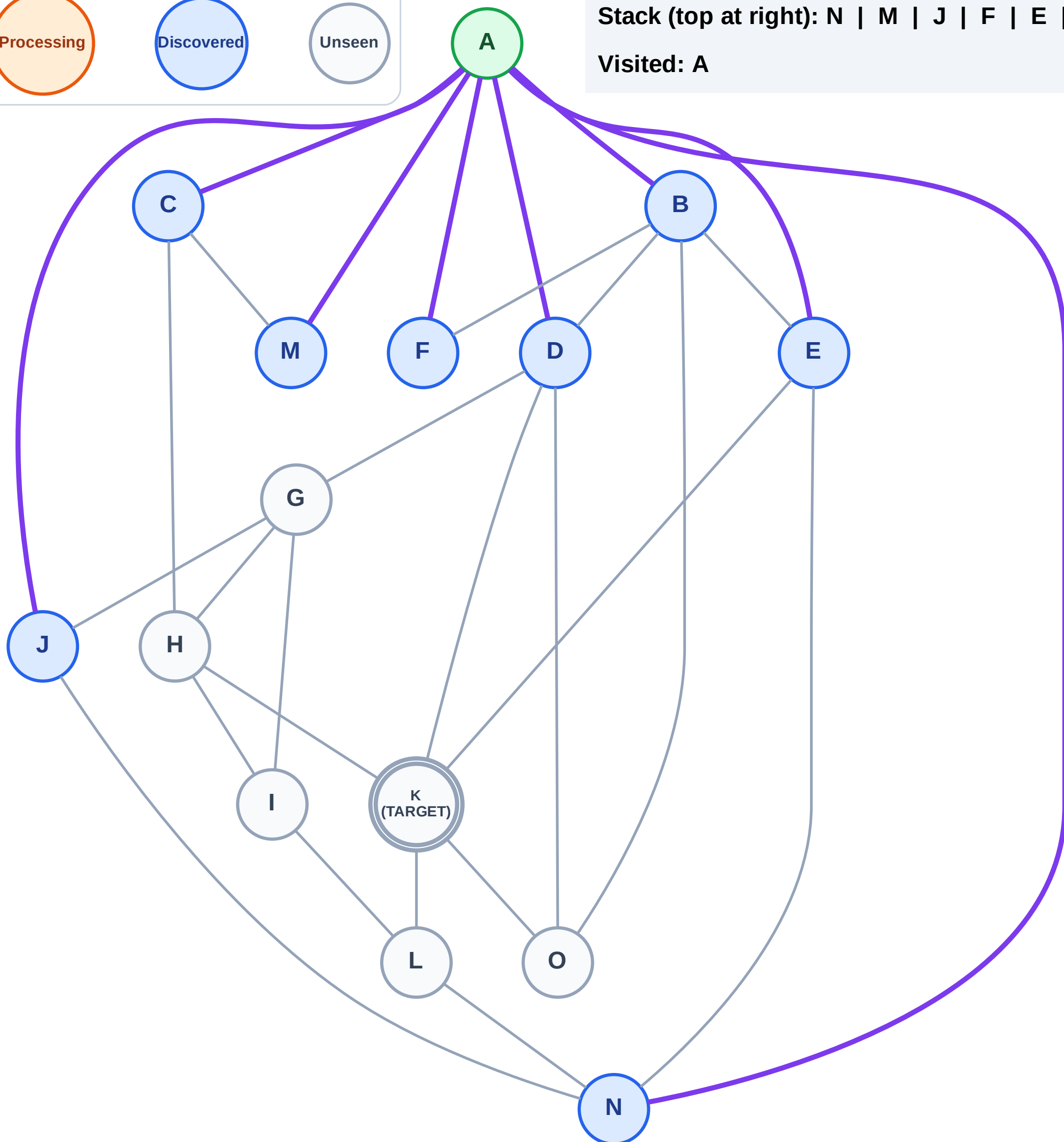
Step 3: Discover N, M, J, F, E, D, C, B from A

Legend



Stack (top at right): N | M | J | F | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

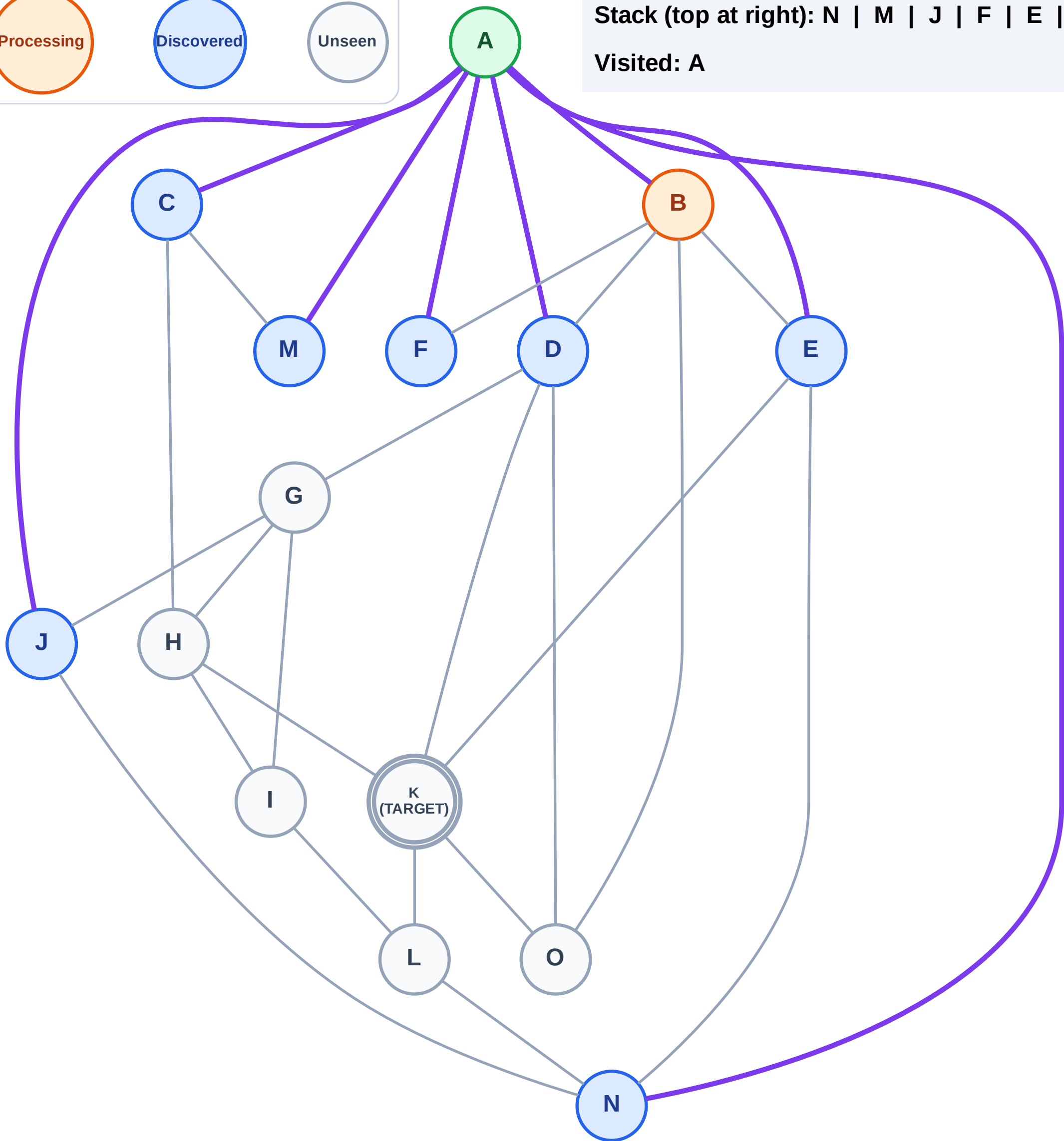
Processing

Discovered

Unseen

Stack (top at right): N | M | J | F | E | D | C

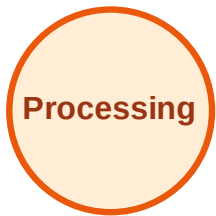
Visited: A



DFS Traversal

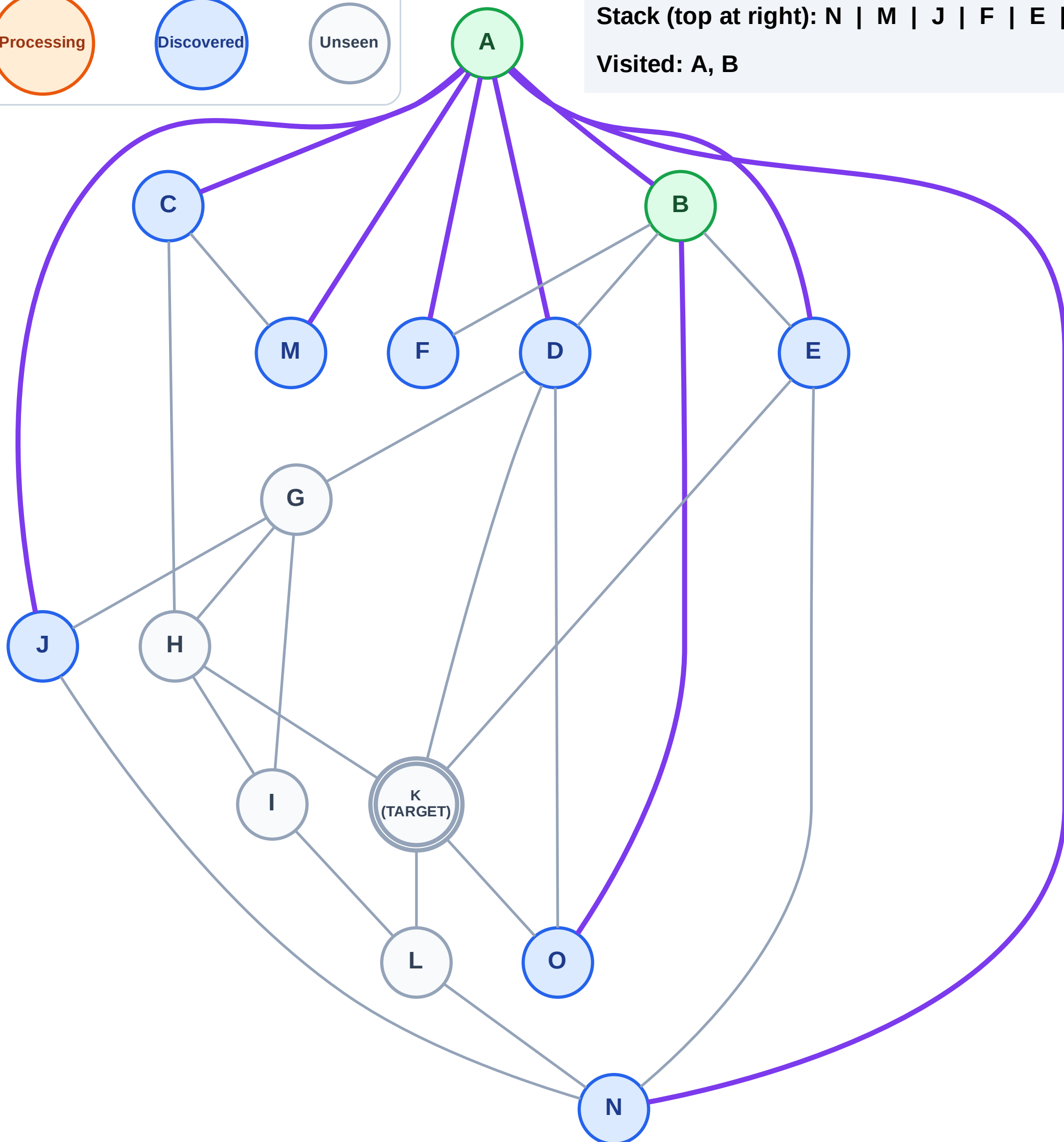
Step 5: Discover O from B

Legend

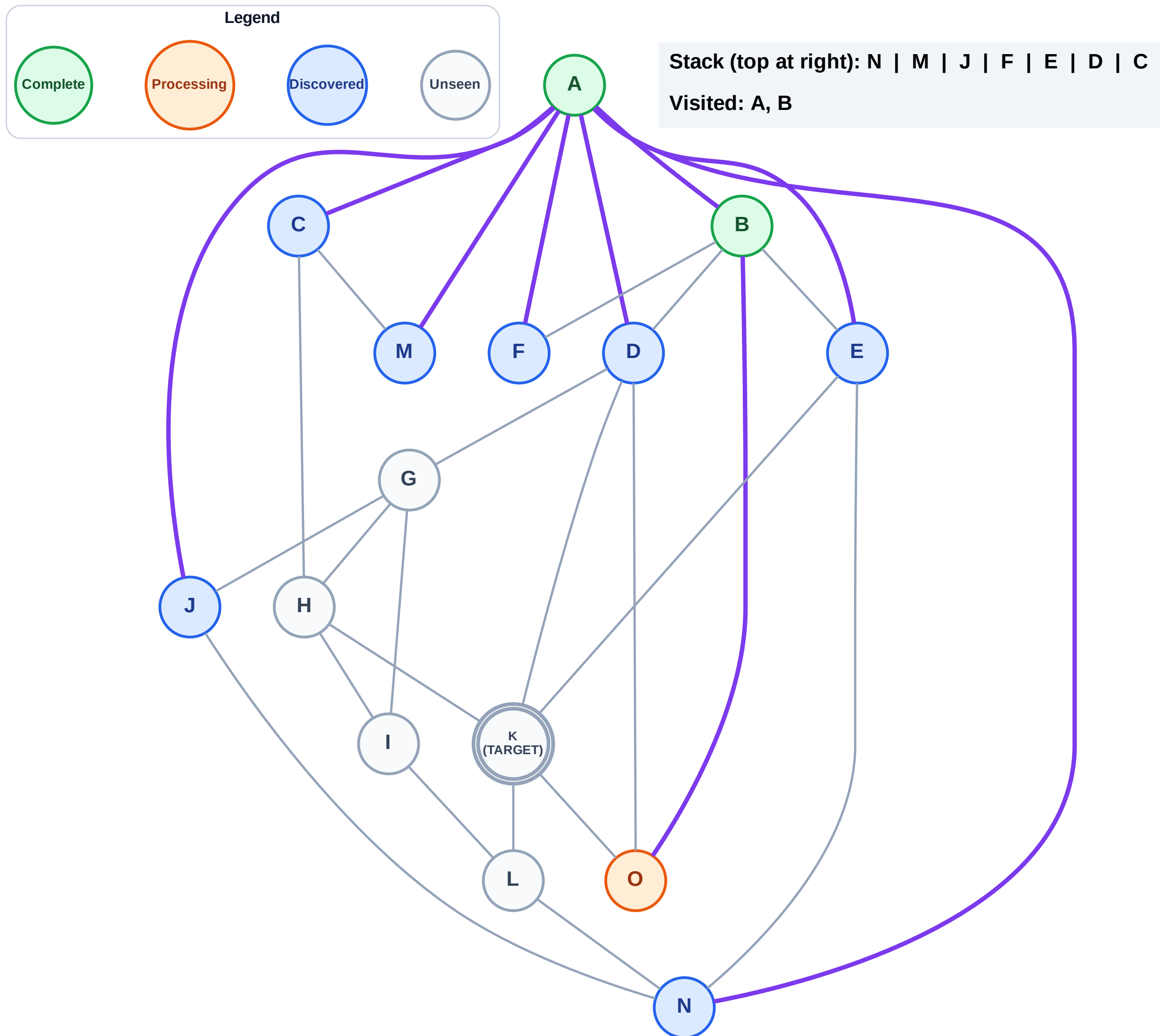


Stack (top at right): N | M | J | F | E | D | C | O

Visited: A, B



Step 6: Process node O



DFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (3 hops): A → B → O → K

